Groups and Lie Algebras: Exercise Sheet 8

Clebsch–Gordan Coefficients for $\mathfrak{sl}(2)$ In this exercise we will be using the physics convention of indexing representations by spin which is half of the weight introduced in the lecture. In this convention the irreducible representation of $\mathfrak{sl}(2)$ of spin j (dimension $2j+1$) is denoted V_j, with standard basis $\{|j,m\rangle : m = -j, -j+1, \dots, j\}$ satisfying

$$\begin{aligned}
H\,|j,m\rangle &= 2m\,|j,m\rangle\,,\\
E\,|j,m\rangle &= \sqrt{(j-m)(j+m+1)}\;|j,m+1\rangle\,,\\
F\,|j,m\rangle &= \sqrt{(j+m)(j-m+1)}\;|j,m-1\rangle\,,
\end{aligned}$$

where H, E, F are the standard generators of $\mathfrak{sl}(2)$ with relations $[H,E] = +2E$, $[H,F] = -2F$ and $[E,F] = H$.

Part 1: The tensor product decomposition

The tensor product $V_j \otimes V_{j'}$ carries a representation of $\mathfrak{sl}(2)$ via

$$X \cdot (v \otimes w) = (Xv) \otimes w + v \otimes (Xw), \qquad X \in \mathfrak{sl}(2).$$

(a) Show that the vectors $|j,m\rangle \otimes |j',m'\rangle$ are eigenvectors of H in $V_j \otimes V_{j'}$, and determine their eigenvalues.

(b) Using the eigenvalue decomposition, confirm that

$$V_j \otimes V_{j'} \;\cong\; V_{j+j'} \oplus V_{j+j'-1} \oplus \cdots \oplus V_{|j-j'|}$$

by verifying that both sides have the same dimension and the same eigenvalue multiplicities for H.

Part 2: Clebsch–Gordan coefficients

By the decomposition above, $V_j \otimes V_{j'}$ admits a second basis $\{|J,M\rangle : J = |j-j'|, \dots, j+j',\; M = -J, \dots, J\}$, where each $|J,M\rangle$ is a highest-weight vector (or descendant thereof) in the copy of V_J inside $V_j \otimes V_{j'}$. The **Clebsch–Gordan coefficients** $\langle j,m;\, j',m' \mid J,M\rangle$ are defined by the change-of-basis expansion

$$|J,M\rangle \;=\; \sum_{\substack{m,m' \\ m+m'=M}} \langle j,m;\, j',m' \mid J,M\rangle\; |j,m\rangle \otimes |j',m'\rangle\,.$$

(a) Explain why only terms with $m + m' = M$ appear in the sum above.

(b) Show that the highest-weight vector $|J_{\max}, J_{\max}\rangle$ with $J_{\max} = j + j'$ must be proportional to $|j,j\rangle \otimes |j',j'\rangle$, and fix the normalisation by choosing this coefficient to be 1.

(c) Starting from $|J_{\max}, J_{\max}\rangle$, explain how the remaining basis vectors $|J_{\max}, M\rangle$ for $M < J_{\max}$ can be obtained by repeated application of F, and how this determines further Clebsch–Gordan coefficients.

(d) Explain how the highest-weight vector for the next summand $V_{J_{\max}-1}$ can be found by requiring it to have eigenvalue $2(J_{\max} - 1)$ under H and to be orthogonal to $|J_{\max},\ J_{\max} - 1\rangle$.

Part 3: Explicit computation for $V_{1/2} \otimes V_{1/2}$
Consider the tensor product $V_{1/2} \otimes V_{1/2}$. Write $|+\rangle := |\frac{1}{2}, \frac{1}{2}\rangle$ and $|-\rangle := |\frac{1}{2}, -\frac{1}{2}\rangle$ for the standard basis of $V_{1/2}$.

(a) Write down the four basis vectors of $V_{1/2} \otimes V_{1/2}$ and their H-eigenvalues.

(b) According to the decomposition of Part 1, we have $V_{1/2} \otimes V_{1/2} \cong V_1 \oplus V_0$. Determine all Clebsch–Gordan coefficients explicitly: find the three vectors $|1,1\rangle$, $|1,0\rangle$, $|1,-1\rangle$ and the singlet $|0,0\rangle$, expressed in the basis $\{|+\rangle \otimes |+\rangle\,, |+\rangle \otimes |-\rangle\,, |-\rangle \otimes |+\rangle\,, |-\rangle \otimes |-\rangle\}$.

(c) Write the 4×4 change-of-basis matrix C between the two bases, with rows indexed by (J, M) and columns by (m, m'). Verify that C is orthogonal.

(d) Verify explicitly that $E\,|0,0\rangle = 0$ and $F\,|1,-1\rangle = 0$ in $V_{1/2} \otimes V_{1/2}$.

Part 4: The case $V_1 \otimes V_{1/2}$

(a) List the H-eigenvalues and their multiplicities in $V_1 \otimes V_{1/2}$.

(b) Determine the Clebsch–Gordan decomposition $V_1 \otimes V_{1/2} \cong V_{3/2} \oplus V_{1/2}$ by finding all six basis vectors $|J, M\rangle$ explicitly. (*Hint:* Start from the highest-weight vector $|\frac{3}{2}, \frac{3}{2}\rangle$ and use E^-; then find $|\frac{1}{2}, \frac{1}{2}\rangle$ by orthogonality.)

(c) Read off the Clebsch–Gordan coefficients for this case and arrange them in a table of the form

(m, m')	$(1, \frac{1}{2})$	$(0, \frac{1}{2})$	$(1, -\frac{1}{2})$	$(-1, \frac{1}{2})$	$(0, -\frac{1}{2})$	$(-1, -\frac{1}{2})$
$\lvert\frac{3}{2}, \frac{3}{2}\rangle$						
$\lvert\frac{3}{2}, \frac{1}{2}\rangle$						
$\lvert\frac{3}{2}, -\frac{1}{2}\rangle$						
$\lvert\frac{3}{2}, -\frac{3}{2}\rangle$						
$\lvert\frac{1}{2}, \frac{1}{2}\rangle$						
$\lvert\frac{1}{2}, -\frac{1}{2}\rangle$						

Part 5: Application – spin-orbit coupling

In quantum mechanics, the state space of a particle with orbital angular momentum $\ell = 1$ and spin $s = \frac{1}{2}$ is $V_1 \otimes V_{1/2}$, where the first factor describes the orbital degrees of freedom and the second the spin.

Recall that the generators H, E, F are related to the standard angular momentum operators by

$$L_z = \tfrac{1}{2}H, \qquad L_+ = E, \qquad L_- = F.$$

The **Casimir operator** of a representation V_J of $\mathfrak{sl}(2)$ is the operator

$$\mathbf{J}^2 := L_z^2 + \tfrac{1}{2}(L_+L_- + L_-L_+) \;=\; \tfrac{1}{4}H^2 + \tfrac{1}{2}(EF + FE),$$

which commutes with all three generators H, E, F. By Schur's lemma it therefore acts as a scalar on any irreducible representation. We write $\mathbf{L}^2$, $\mathbf{S}^2$, and $(\mathbf{L}+\mathbf{S})^2$ for the Casimir operators acting on V_1, $V_{1/2}$, and the combined representation $V_1 \otimes V_{1/2}$ respectively.

The spin-orbit interaction is described by the operator

$$\mathbf{L}\cdot\mathbf{S} \;=\; \frac{1}{2}\big((\mathbf{L}+\mathbf{S})^2 - \mathbf{L}^2 - \mathbf{S}^2\big),$$

which follows from expanding $(\mathbf{L}+\mathbf{S})^2 = \mathbf{L}^2 + 2\,\mathbf{L}\cdot\mathbf{S} + \mathbf{S}^2$ and solving for $\mathbf{L}\cdot\mathbf{S}$.

(a) Show that the Casimir operator acts on V_J as the scalar $J(J+1)\,\mathrm{id}$. What are the actions of $\mathbf{L}^2$ on V_1 and $\mathbf{S}^2$ on $V_{1/2}$?

(b) Using the decomposition $V_1 \otimes V_{1/2} \cong V_{3/2} \oplus V_{1/2}$ from Part 4, show that $\mathbf{L}\cdot\mathbf{S}$ acts as a scalar on each summand, and determine these scalars. (*Hint:* Apply the identity above, using the fact that $(\mathbf{L}+\mathbf{S})^2$ acts as $J(J+1)\,\mathrm{id}$ on V_J.)

(c) The spin-orbit Hamiltonian is $\mathcal{H} = \lambda\,\mathbf{L}\cdot\mathbf{S}$ for a real constant $\lambda > 0$. Using your result from (b), write down the energy eigenvalues of $\mathcal{H}$ on $V_1 \otimes V_{1/2}$ and their degeneracies.

(d) The *good quantum numbers* for the spin-orbit interaction are the eigenvalues of $(\mathbf{L}+\mathbf{S})^2$ and $L_z + S_z$, rather than those of L_z and S_z separately. Explain, using the decomposition of Part 4, why the basis $\{|J, M\rangle\}$ diagonalises $\mathcal{H}$, while the tensor product basis $\{|1, m\rangle \otimes |\frac{1}{2}, m'\rangle\}$ does not.

(e) Using the explicit vectors from Part 4(b), verify your eigenvalue computation from Part (b) directly for the two vectors $|\frac{3}{2}, \frac{1}{2}\rangle$ and $|\frac{1}{2}, \frac{1}{2}\rangle$ by acting with $\mathbf{L}\cdot\mathbf{S}$ in the tensor product basis. (*Hint:* Write $\mathbf{L}\cdot\mathbf{S} = \frac{1}{2}(L_+S_- + L_-S_+) + L_zS_z$ and act term by term, where $L_\pm, S_\pm$ act on the respective factors.)